

Time & Work — Question Pattern Catalog

Every High-Yield Phrasing, Grouped by the Trick That Solves It

Aptitude Grind

Time & Work questions look varied, but almost every one of them reduces to a handful of tricks from `02-fast-tricks-and-shortcuts.tex`. This file catalogues the actual **phrasings** you'll meet in TCS NQT (Ninja + Digital) and similar placement tests, grouped by category, each with the fastest method and a verified mini-example. Skim this before a test to pattern-match a question to its solution method in seconds.

Category A — Two (or More) People Working Together

A1. Direct Combined-Work Question

Phrasing

“A can finish a work in 6 days, B in 12 days. In how many days can they finish it together?”

Method: $xy/(x+y)$ from `01-fundamentals.tex` §3.

Answer: $(6 \times 12)/(6 + 12) = 4$ **days**.

A2. Three-or-More-Workers Combined Question

Phrasing

“A, B, and C can individually finish a job in 12, 15, and 20 days. Working together, how long will they take?”

Method: LCM Method (`02-fast-tricks-and-shortcuts.tex` §1). $\text{LCM}(12,15,20) = 60 \Rightarrow$ rates 5, 4, 3 units/day $\Rightarrow$ combined rate = 12 units/day $\Rightarrow$ **5 days**.

Category B — Efficiency Ratio Problems

B1. “X Times as Efficient” Question

Phrasing

“A is thrice as efficient as B. B alone can complete a work in 24 days. Find the time taken by A alone, and by A and B together.”

Method: Rate-First Thinking (§2). B's rate = $1/24$; A's rate = $3/24 = 1/8 \Rightarrow$ A alone = **8 days**; together = $(8 \times 24)/(8 + 24) = 6$ **days**.

B2. Time-Ratio-to-Efficiency-Ratio Conversion

Phrasing

“The ratio of the time taken by P and Q to complete a task is 4:7. What is the ratio of their efficiencies?”

Method: Efficiency and time are always inversely proportional (01-fundamentals.tex §6) $\Rightarrow$ efficiency ratio = **7:4**.

Category C — Worker Joining or Leaving Midway

C1. Someone Joins Partway Through

Phrasing

“A can complete a job alone in 18 days. A works alone for 6 days, then B (who alone takes 9 days) joins. How many total days does the job take?”

Method: Completed-work-then-remaining-work split (02-fast-tricks-and-shortcuts.tex §4). A’s 6-day share = $6/18 = 1/3$; remaining = $2/3$; combined rate = $1/18 + 1/9 = 1/6$; extra days needed = $(2/3)/(1/6) = 4$. **Total = 6 + 4 = 10 days.**

C2. Workers Leave Partway Through

Phrasing

“24 workers can finish a job in 18 days. After 6 days, 6 workers leave. In how many more days will the remaining workers finish the job?”

Method: Worker-Day Method (02-fast-tricks-and-shortcuts.tex §3). Total = $24 \times 18 = 432$ worker-days; done = $24 \times 6 = 144$; remaining = 288; workers left = 18 $\Rightarrow$ **16 more days.**

Category D — Men / Women / Children Equivalence

D1. Given an Equivalence Ratio, Find a Mixed-Group Time

Phrasing

“2 men do the same amount of work as 3 women. 2 men alone can finish a job in 12 days. In how many days can 4 men and 6 women together finish the same job?”

Method: Convert to a common rate unit first (01-fundamentals.tex §7). 2 men = 12 days $\Rightarrow$ 1 man = $1/24$; since 2 men = 3 women in output, 1 woman = $1/36$. Combined rate of 4 men + 6 women = $4/24 + 6/36 = 1/6 + 1/6 = 1/3 \Rightarrow$ **3 days.**

D2. Men/Women/Children Together

Phrasing

“If 2 men or 4 women or 6 children can complete a job in the same time, how many days will 1 man, 2 women, and 3 children together take, given 2 men alone take 12 days?”

Method: Same conversion approach as D1 — every category is first expressed in terms of a single “man-unit,” then rates are added directly.

Category E — Pipes and Cisterns

E1. Fill and Empty Pipe Together

Phrasing

“Pipe A can fill a tank in 8 hours; pipe B can empty it in 12 hours. If both are opened together, how long will the tank take to fill?”

Method: Net rate (01-fundamentals.tex §9). $1/8 - 1/12 = 1/24 \Rightarrow$ **24 hours.**

E2. A Pipe Is Closed Partway Through

Phrasing

“Fill pipe A takes 10 hours, outlet pipe B takes 15 hours. Both are opened together, but B is closed after 3 hours. How long does the tank take to fill in total?”

Method: Two-phase split (02-fast-tricks-and-shortcuts.tex §8). Net rate while both open = $1/10 - 1/15 = 1/30$; work in 3 hr = $1/10$; remaining $9/10$ filled by A alone at $1/10/\text{hr} = 9$ more hours $\Rightarrow$ **Total = 12 hours.**

E3. Multiple Fill Pipes, One Outlet

Phrasing

“Three pipes P, Q, R can fill a tank in 12, 15, and 20 minutes respectively, while an outlet pipe S can empty it in 10 minutes. If all four are opened together, will the tank ever fill?”

Method: Sum every inlet rate, subtract the outlet rate (01-fundamentals.tex §9). If the net rate is zero or negative, the tank **never fills** — always check the sign of the net rate before computing a time.

Category F — Alternate-Day Working

F1. Two Workers Alternate Days

Phrasing

“A can complete a job in 10 days, B in 15 days. Working on alternate days, starting with A, in how many days will the job be completed?”

Method: Cycle-first method (`02-fast-tricks-and-shortcuts.tex` §5). 2-day cycle work = $1/10 + 1/15 = 1/6$; since $1/6 \times 6 = 1$ exactly, the job finishes in exactly **12 days** (6 full cycles), with **no partial final day**.

F2. Alternate Days With a Leftover Fraction

Phrasing

“A can complete a job in 4 days, B in 6 days. Working on alternate days starting with A, how many full days pass before the job is finished, and who finishes it?”

Method: Same cycle-first method, but here the cycle work ($1/4 + 1/6 = 5/12$) does **not** divide evenly into 1 — find how many full cycles fit, then solve the small leftover fraction with whichever worker’s turn comes next (don’t assume it always lands on a clean day).

Category G — Wages Based on Work

G1. Split Payment by Work Contributed

Phrasing

“A (10 days alone) and B (15 days alone) work together on a job and are jointly paid Rs. 1500. What is each person’s share?”

Method: Wages-by-work-share (`02-fast-tricks-and-shortcuts.tex` §7). Rate ratio $A : B = 3 : 2 \Rightarrow A$ gets $1500 \times 3/5 = \text{Rs. } 900$, B gets $1500 \times 2/5 = \text{Rs. } 600$.

G2. Three-Way Wage Split With a Helper

Phrasing

“A and B can together do a job in 10 days. With the help of C, they finish it in 6 days. If the total payment is Rs. 3000, how much should C receive?”

Method: Find C’s rate by subtraction (difference-formula logic from `01-fundamentals.tex` §5, applied to a three-person combined rate instead of two), then split wages by rate ratio exactly as in G1.

Category H — Machine / Printer / Robot Work

H1. Direct Machine-Rate Question

Phrasing

“Printer A can print 1000 pages in 5 hours, Printer B can print the same 1000 pages in 8 hours. If both print together, how long will it take?”

Method: Identical to a two-worker combined-work question (01-fundamentals.tex §8) — a machine’s rate is treated exactly like a person’s rate. $(5 \times 8)/(5 + 8) = 40/13$ hours.

H2. Machine Stopped Partway

Phrasing

“Machine P can complete a task in 8 hours, machine Q in 10 hours, machine R in 12 hours. All three start together, but P is shut off after 2 hours. In approximately how much more time will Q and R finish the rest?”

Method: Same two-phase split as E2 — compute work done by all three in the first phase, subtract from 1, then solve the remainder using only the machines still running.

Category I — Percentage / Fractional Work Completed

I1. Given a Percentage, Find the Full-Job Time

Phrasing

“A can complete 75% of a work in 15 days. In how many days can A alone complete the entire work? B then finishes the remaining 25% in 4 days — how many days would B alone take for the full job?”

Method: Scale up directly — if 75% takes 15 days, 100% takes $15/0.75 = 20$ days for A. Similarly, B’s 25% in 4 days scales to $4/0.25 = 16$ days for the full job.

I2. Fractional Work Split Between Two Workers

Phrasing

“A can do 40% of a job in a certain time; B does the remaining 60%. If A takes 8 days for their share, and A and B have the same efficiency, how long does B take for their share?”

Method: Convert percentages to a work ratio ($40 : 60 = 2 : 3$), then scale A’s time by that ratio to find B’s time, since equal efficiency means time is proportional to the fraction of work assigned.

Category J — Variable Efficiency (Speed Changes Mid-Job)

J1. Efficiency Doubles/Triples Partway

Phrasing

“Working at a constant rate, A can finish a job in 10 days. A works at the normal rate for the first 4 days, then works at double efficiency for the rest. How many total days does the job take?”

Method: Treat this as two separate phases with two different rates, exactly like the two-phase pipe/machine split in E2/H2. First 4 days at rate $1/10$ complete $2/5$; remaining $3/5$ at double rate ($1/5/\text{day}$) takes 3 more days $\Rightarrow$ **Total = 7 days.**

J2. Efficiency Changes by a Given Percentage

Phrasing

“A normally takes 20 days for a job. If A’s efficiency increases by 25% starting from day one, how many days will A now take?”

Method: Percentage-to-time formula (02-fast-tricks-and-shortcuts.tex §6). New Time = $20 \times 100/125 = 16$ **days.**

Priority Order for Limited Prep Time

If you’re short on time, drill these first — they cover the large majority of what actually appears in TCS-style placement tests, roughly in order of frequency:

Priority List

1. **A1** — Two people working together (the base formula everything else builds on)
2. **B1** — Efficiency ratio (“X times as efficient”)
3. **C1/C2** — Someone joins or leaves midway
4. **D1** — Men/Women/Children equivalence
5. **E1/E2** — Pipes and cisterns, including a pipe closing partway
6. **F1** — Alternate-day working
7. **G1** — Wages split by work share
8. **H1** — Machine/Printer/Robot work (same math as A1, different wrapper)
9. **I1** — Percentage of work completed
10. **J1/J2** — Variable efficiency (lower frequency, but appears occasionally in TCS Digital)

These ten patterns, combined with the LCM Method and rate-first thinking from 02-fast-tricks-and-shortcuts, cover almost everything TCS typically asks while avoiding the low-return, highly complex worker puzzles seen in exams like CAT or SSC CGL.